Linear Equations.

CHAPTER

11

Linear Partial Differential Equations

11.1 Homogeneous partial differential equations

An equation of the form

$$\frac{\partial^n z}{\partial x^n} + k_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + \cdots\cdots + k_n \frac{\partial^n z}{\partial y^n} = F(x, y) \qquad (11.1)$$

where $k_1, k_1, \cdots\cdots, k_n$ are constants, is called a linear homogeneous partial differential equation of order n with constant coefficients. Equation (11.1) can be written in symbolic form as

$$(D^n + k_1 D^{n-1} D' + \cdots\cdots + k_n D'^n) z = F(x, y) \qquad (11.2)$$

or $$(D_x^n + k_1 D_x^{n-1} D_y + \cdots\cdots + k_n D_y^n) z = F(x, y)$$

or $$f(D_x, D_y) z = F(x, y)$$

where $$D_x^n = \frac{\partial^n}{\partial x^n}, \quad D_y^n = \frac{\partial^n}{\partial y^n}.$$

Like ordinary linear differential equations with constant coefficients, the complete solution of equation (11.1) consists of the complementary function (C. F.) and particular integral (P. I). The complementary function is the complete solution of $f(D_x, D_y) z = 0$ which contains n arbitrary constants. The particular integral is the particular solution of equation (11.2).

11.2 Methods of finding complementary functions

Let $F(D_x, D_y) z = (D_x - m_1 D_y)(D_x - m_2 D_y) \cdots\cdots (D_x - m_n D_y) z$

where $$m_1, m_2, \cdots\cdots, m_n$$

are real and distinct.

Consider $(D_x - m_r D_y)z = 0.$

or $\frac{\partial z}{\partial x} - m_r \frac{\partial z}{\partial y} = 0$ or, $p - m_r q = 0$

which is a Lagrange's linear equation. The subsidiary equations are

$$\frac{dx}{1} = \frac{dy}{-m_r} = \frac{dz}{0}$$

First two relations give $dy + m_r dx = 0$

$\therefore$ $y + m_r x = c_1$

and $\frac{dz}{0} = \frac{dx}{1}$ yielding $z = c_2$.

Hence $z = \phi_r(y + m_r x)$

is a solution of $(D_x - m_r D_y)z = 0$

where ϕ_r is an arbitrary function.

Hence if $m_1, m_2, \cdots\cdots, m_n$

are all distinct the complete solution is

$$z = \phi_1(y + m_1 x) + \phi_2(y + m_2 x) + \cdots\cdots + \phi_n(y + m_n x)$$

Cor: Let $z = \phi(y + mx)$

be a solution of $F(D_x, D_y)z = 0$

then $D_x^r z = m^r \phi^{(r)}(y + mx)$

and $D_y^r z = \phi^{(r)}(y + mx).$

Thus $F(D_x, D_y)z = (A_0 D^n + A_1 D^{n-1} D_y + A_n D_y^n)z$

$$= (A_0 m^n + A_1 m^{n-1} + A_n)\phi(y + mx)$$

The equation $A_0 m^n + A_1 m^{n-1} + \cdots\cdots + A_n = 0$

is called the auxiliary equation. The auxiliary equation is simply obtained by replacing $D_x z$ by m and $D_y z$ by 1 in

$$F(D_x, D_y)z = 0.$$

Example 11.1 Solve $2\frac{\partial^2 z}{\partial x^2} + 5\frac{\partial^2 z}{\partial x \partial y} + 2\frac{\partial^2 z}{\partial y^2} = 0$

Solution The given equation can be written as

$$\left(2D_x^2 + 5D_x D_y + 2D_y^2\right)z = 0.$$

The auxiliary equation $2m^2 + 5m + 2 = 0$

has the roots $m = -2, -\frac{1}{2}.$

Hence the complete solution is

$$z = \phi_1(y - 2x) + \phi_2\left(y - \frac{1}{2}x\right).$$

Example 11.2 Solve $\frac{\partial^4 z}{\partial x^4} - \frac{\partial^4 z}{\partial y^4} = 0.$

Solution the equation is $\left(D_x^4 - D_y^4\right)z = 0.$

The auxiliary equation is $m^4 - 1 = 0$ giving $m = \pm 1, \pm i.$

Thus the complete solution is

$$z = \phi_1(y + x) + \phi_2(y - x) + \phi_3(y + ix) + \phi_4(y - ix).$$

11.3 For repeated roots of auxiliary equation

If the roots of the auxiliary equation are equal say m then

$$(D_x - mD_y)(D_x - mD_y)z = 0 \quad (11.3)$$

Putting $(D_x - mD_y)z = u$

in (11.3), we have $(D_x - mD_y)u = 0$

which gives $u = \phi_1(y + mx)$.

Thus equation (11.3) becomes

$$(D_x - mD_y)z = \phi_1(y + mx)$$

or $p - mq = \phi_1(y + mx)$

This is again Lagrange's linear equation and the subsidiary equations are

$$\frac{dx}{1}=\frac{dy}{-m}=\frac{dz}{\phi_1(y+mx)}$$

which gives $$y+mx=a$$

and $$dz=\phi_1(a)dx$$

i.e. $$z=x\phi_1(a)+b.$$

Therefore the complete solution of (11.3) is

$$z=x\phi_1(y+mx)+\phi_2(y+mx)$$

$\therefore$ If $(D_x-mD_y)z=0=(D_x-mD_y)^k=0$

then $$z=\phi_1(y+mx)+x\phi_2(y+mx)+x^2\phi_3(y+mx)$$
$$+\cdots\cdots x^{k-1}\phi_k(y+mx)$$

Example 11.3 Solve $r+8s+16t=0.$

Solution The equation can be write as

$$\left(D_x^2+8D_xD_y+16D_y^2\right)z=0.$$

The auxiliary equation $m^2+8m+16=0$

has two equal roots $m=-4,\ -4$

and the complete solution is $z=\phi_1(y-4x)+x\phi_2(y-4x).$

Particular integral

To find the particular integral, consider the second order partial differential equation

$$\left(D_x^2+k_1D_xD_y+k_2D_y^2\right)z=F(x,\ y)$$

or $$f\left(D_x,\ D_y\right)z=F(x,\ y).$$

Particular integral $=\dfrac{1}{f\left(D_x,\ D_y\right)}F(x,\ y).$

We have the following 4 cases.

Case(a): When $F(x,\ y)=e^{ax+by}$.

$$D_xe^{ax+by}=ae^{ax+by},\ D_y^a e^{ax+by}=be^{ax+by}$$

$$D_x^2e^{ax+by}=a^2e^{ax+by},\ D_y^2e^{ax+by}=b^2e^{ax+by}$$

and $$D_xD_ye^{ax+by}=abe^{ax+by}$$

Thus $$\left(D_x^2+k_1D_xD_y+k_2D_y^2\right)e^{ax+by}=\left(a^2+k_1ab+k_2b^2\right)e^{ax+by}$$

or $$f(D_x,\ D_y)e^{ax+by}=f(a,\ b)e^{ax+by}$$

Operating both sides by $\dfrac{1}{f(D_x,\ D_y)}$

we get $$\text{P. I}=\frac{1}{f(D_x,\ D_y)}F(x,\ y)=\frac{1}{f(a,\ b)}e^{ax+by},$$

provided $$f(a,\ b)\neq 0.$$

Working Rule to evaluate $\dfrac{1}{f(D_x,\ D_y)}e^{ax+by}$ when $f(a,\ b)=0$

(i) Differentiate $f(D_x,\ D_y)$

with respect to D_x partially and multiply the expression by x, so that

$$\frac{1}{f(D_x,\ D_y)}e^{ax+by}=x\frac{1}{f'(D_x,\ D_y)}e^{ax+by}.$$

(ii) If $f'(a,\ b)$ is also zero, differentiate $f'(D_x,\ D_y)$

with respect to D_x partially and multiply by x again so that

$$\frac{1}{f(D_x,\ D_y)}e^{ax+by}=x^2\frac{1}{f''(D_x,\ D_y)}e^{ax+by}.$$

Proceed differentiating in this way every time and multiplying by x as long as the derivative of $f(D_x,\ D_y)$ vanishes when $D_x=a$ and $D_y=b$.

Case(b): When $f(x, y)$ be of the form $\begin{smallmatrix}\sin\\ \cos\end{smallmatrix}(ax+by)$

Here $$P.I = \frac{1}{f\left(D_x^2,\, D_xD_y,\, D_y^2\right)} \begin{smallmatrix}\sin\\ \cos\end{smallmatrix}(ax+by)$$

$$= \frac{1}{f\left(-a^2, -ab, -b^2\right)} \begin{smallmatrix}\sin\\ \cos\end{smallmatrix}(ax+by)$$

provided $$f\left(-a^2, -ab, -b^2\right) \neq 0.$$

If $$f\left(-a^2, -ab, -b^2\right) = 0$$

then P. I. is obtained in similar way as in the case of

$$\frac{1}{f(D_x, D_y)} e^{ax+by}.$$

Case(c): When $f(x, y) = x^a y^b$, a, b are constants.

Here $$P.I = \frac{1}{f\left(D_x, D_y\right)} x^a y^b = \left[f\left(D_x, D_y\right)\right]^{-1} x^a y^b$$

and to evaluate it expand $\left[f\left(D_x, D_y\right)\right]^{-1}$

in ascending power of D_x or D_y using the binomial theorem and then operate on $x^a y^b$ term by term.

Case (d): When $f(x, y)$ is any function of x and y.

Here $$P.I = \frac{1}{f\left(D_x, D_y\right)} f(x, y)$$

and to evaluate it, resolve $\dfrac{1}{f\left(D_x, D_y\right)}$

into partial fractions considering $f\left(D_x, D_y\right)$

as a function of D_x only and operate each partial fraction on $f(x, y)$ noting that

$$\frac{1}{D_x - mD_y} f(x, y) = \int f(x, c - mx)\, dx$$

where c is replaced by $y + mx$ after integration.

Example 11.4 Solve $\left(D_x^2 - D_xD_y - 2D_y^2\right)z = 2x + 3y.$

Solution The auxiliary equation is $m^2 - m - 2 = 0$ giving $m = 2, -1.$

Hence the complementary function is

$$z_c = \phi_1(y + 2x) + \phi_2(y - x)$$

We now proceed to find particular integral.

First method $$z_p = \frac{1}{\left(D_x^2 - D_xD_y - 2D_y^2\right)}(2x + 3y)$$

$$= \frac{1}{D_x^2} \cdot \frac{1}{1 - \dfrac{D_y}{D_x} - \dfrac{2D_y^2}{D_x^2}}(2x + 3y)$$

$$= \frac{1}{D_x^2}\left[\left\{1 + \frac{D_y}{D_x} + \cdots\cdots\right\}(2x + 3y)\right]$$

$$= \frac{1}{D_x^2}[2x + 3y + 3x] = \frac{1}{D_x^2}[5x + 3y]$$

$$= \frac{5}{6}x^3 + \frac{3x^2y}{2}$$

Now $$z = z_c + z_p$$

Second method $$z_p = \frac{1}{\left(D_x^2 - D_xD_y - 2D_y^2\right)}(2x + 3y)$$

$$= \frac{1}{\left(D_x - 2D_y\right)}\left[\frac{1}{D_x + D_y}(2x + 3y)\right]$$

Put $$u = \frac{1}{D_x + D_y}(2x + 3y)$$

i.e. $$(D_x + D_y)u = 2x + 3y$$

Auxiliary equation is $$\frac{dx}{1} = \frac{dy}{1} = \frac{du}{2x + 3y}$$

giving $$y - x = a$$

or $$y = x + a$$

Then $$\frac{dx}{1} = \frac{du}{5x + 3a}.$$

On integrating we get $u = \dfrac{5x^2}{2} + 3ax = \dfrac{5x^2}{2} + 3x(y - x)$

$$= 3xy - \frac{x^2}{2}.$$

Now $$z_p = \frac{1}{(D_x - 2D_y)}\left(3xy - \frac{x^2}{2}\right)$$

i.e. $$(D_x - 2D_y)z_p = 3xy - \frac{x^2}{2}$$

Auxiliary equation is $\dfrac{dx}{1} = \dfrac{dy}{-2} = \dfrac{dz_p}{3xy - \frac{x^2}{2}}.$

From the first two ratios we get $y + 2x = b$

i.e. $$y = b - 2x.$$

Then $$\frac{dx}{1} = \frac{dz_p}{3bx - \left(\frac{13}{2}x^2\right)}$$

$$\therefore \quad z_p = \frac{3bx^2}{2} - \frac{13x^3}{6}$$

$$= \frac{3}{2}x^2(y + 2x) - \frac{13x^3}{6}$$

$$= \frac{5x^3}{6} + \frac{3}{2}x^2 y.$$

$$\therefore \quad z = z_c + z_p$$

Example 11.5 Solve $(D_x^2 + 4D_x D_y + 4D_y^2)z = e^{2x+y}$.

Solution The auxiliary equation is $m^2 + 4m + 4 = 0$

giving $$m = -2, -2.$$

Hence $$C.F = \phi_1(y - 2x) + x\phi_2(y - 2x)$$

Also $$P.I = \frac{1}{D_x^2 + 4D_x D_y + 4D_y^2} e^{2x+y}.$$

Here e^{2x+y} is a function of the form $\phi(ax + by)$

and $$D_x^2 + 4D_x D_y + 4D_y^2$$

is a homogeneous function of D_x, D_y of degree 2.

Integrating e^{2x+y} twice with respect to $2x + y$, we get $\dfrac{e^{2x+y}}{4}$.

Also putting 2 for D_x and 1 for D_y

$$\text{P.I} = \frac{1}{4 + 4.2.1 + 4.1^2} e^{2x+y} = \frac{e^{2x+y}}{16}.$$

Therefore the complete solution is

$$z = \phi_1(y - 2x) + x\phi_2(y - 2x) + \frac{1}{16}e^{2x+y}.$$

Example 11.6 Solve $(D_x^3 - 2D_x^2 D_y - D_x D_y^2 + 2D_y^3)z = e^{x+y}$.

Solution The auxiliary equation is

$$m^3 - 2m^2 - m + 2 = 0$$

giving $$m = 1, -1, 2$$

$$\therefore \quad C.F. = \phi_1(y + x) + \phi_2(y - x) + \phi_3(y + 2x)$$

$$P.I = \frac{1}{D_x^3 - 2D_x^2 D_y - D_x D_y^2 + 2D_y^3} e^{x+y}.$$

Here denominator vanishes where

$$D_x = 1, \; D_y = 1$$

$$\therefore \text{P. I} = x\frac{1}{3D_x^2 - 4D_xD_y - D_y^2}e^{x+y}$$

$$= x.\frac{1}{3.1 - 4.1 - 1}e^{x+y} = -\frac{1}{2}xe^{x+y}$$

The complete solution is

$$z = \phi_1(y+x) + \phi_2(y-x) + \phi_3(y+2x) - \frac{x}{2}e^{x+y}$$

Example 11.7 Solve $(D_x - D_y)^2 z = x + \phi(x+y)$.

Solution The auxiliary equation is

$$(m-1)^2 = 0, \text{ giving } m = 1, 1$$

$$\therefore \quad \text{C. F.} = \phi_1(y+x) + x\phi_2(y+x)$$

$$\text{P. I.} = \frac{1}{(D_x - D_y)^2}x[(x+\phi)(x+y)]$$

$$= \frac{1}{(D_x - D_y)^2}x + \frac{1}{(D_x - D_y)^2}\phi(x+y)$$

$$= \frac{1}{D_x^2(1-D_y)^2}x + \frac{x}{2(D_x - D_y)}\phi(x+y)$$

$$= \frac{x^3}{6} + \frac{x^2}{2}\phi(x+y)$$

$\therefore$ The complete solution is

$$z = \phi_2(y+x) + x\phi_2(y+x) + \frac{x^3}{6} + \frac{x^2}{2}\phi(x+y)$$

Example 11.8 Solve $(D_x^2 - 5D_xD_y + 4D_y^2)z = \sin(2x+y)$.

Solution The roots of auxiliary equation are

$$m = 1, 4$$

$$\therefore \quad \text{C. F.} = \phi_1(y+x) + \phi_2(y+4x)$$

$$\text{P. I.} = \frac{1}{D_x^2 - 5D_xD_y + 4D_y^2)}\sin(2x+y)$$

$$= \frac{1}{-4+10-4}\sin(2x+y)$$

$$= \frac{1}{2}\sin(2x+y).$$

The solution is $z = \phi_1(y+x) + \phi_2(y+4x) + \frac{1}{2}\sin(2x+y)$.

Example 11.9 Solve $(D_x^3 - 4D_x^2D_y + 4D_xD_y^2)z = 4\cos(2x+y)$.

Solution The auxiliary equation is $m^3 - 4m^2 + 4m = 0$

giving $$m = 0, 2, 2$$

$$\therefore \quad \text{C. F.} = \phi_1(y) + \phi_2(y+2x) + x\phi_3(y+2x)$$

$$\text{P. I.} = \frac{1}{(D_x^3 - 4D_x^2D_y + 4D_xD_y^2)}.4\cos(2x+y)$$

The denominator becomes zero when $D_x = 2,\ D_y = 1$.

Differentiating denominator w.r. to D_x and multiplying numerator by x, we get

$$\text{P. I.} = \frac{x}{3D_x^2 - 8D_xD_y + 4D_y^2)}4\cos(2x+y)$$

The denominator again vanishes when $D_x = 2,\ D_y = 1$.

Therefore, again differentiating the denominator with respect to D_x and multiplying numerator by x we get

$$\text{P. I.} = x^2\frac{1}{6D_x - 8D_y}4\cos(2x+y)$$

The denominator is of order 1 and does not vanish when

$$D_x = 2,\ D_y = 1.$$

$$\therefore \quad \text{P. I.} = \frac{4x^2}{6.2 - 8.1}\int \cos t\, dt$$

when $\qquad t = 2x + y$

$$= x^2 \sin t = x^2 \sin(2x + y)$$

$$\therefore z = \phi_1(y) + \phi_2(y + 2x) + x\phi_3(y + 2x) + x^2 \sin(2x + y).$$

Example-11.10 Solve $(D_x^2 - D_x D_y - 6D_y^2)z = e^{2y} \sin 3x.$

Solution C. F. $= \phi_1(y + 2x) + \phi_2(y - 3x)$

$$\text{P. I.} = \frac{1}{D_x^2 - D_x D_y - 6D_y^2} e^{2y} \sin 3x$$

$$= e^{2y} \frac{1}{D_x^2 - D_x(D_y + 2) - 6(D_y + 2)^2} \sin 3x$$

$$= e^{2y} \frac{1}{D_x^2 - D_x D_y - 2D_x - 6D_y^2 - 24D_y - 24} \sin 3x$$

$$= e^{2y} \frac{1}{-9 - 2D_x - 24D_y - 24} \sin 3x$$

$$= -e^{2y} \frac{1}{2D_x + 24D_y + 33} \sin 3x$$

$$= -e^{2y} \frac{2D_x + 24D_y - 33}{(2D_x + 24D_y)^2 - 33^2} \sin 3x$$

$$= -e^{2y} \frac{6\cos 3x - 33\sin 3x}{4(-9) - 33^2}$$

$$= e^{2y} \frac{6\cos 3x - 33\sin 3x}{1125}$$

$$= e^{2y} \frac{2\cos 3x - 11\sin 3x}{375}$$

General solution is

$$z = \phi_1(y + 2x) + \phi_2(y - 3x) + e^{2y} \frac{2\cos 3x - 11\sin 3x}{375}.$$

Example-11.11 Solve $(D_x^3 - 4D_x^2 D_y + 5D_x D_y^2 - 2D_y^3)z = (y + x)^{\frac{1}{2}}.$

Solution The auxiliary equation is $m^3 - 4m^2 + 5m - 2 = 0$

giving $\qquad m = 2, 1, 1$

$$\text{C. F.} = \phi_1(y + 2x) + \phi_2(y + x) + x\phi_3(y + x)$$

$$\text{P. I.} = \frac{1}{(D_x^3 - 4D_x^2 D_y + 5D_x D_y^2 - 2D_y^3)}(y + x)^{\frac{1}{2}}$$

$$= \frac{1}{(D_x - D_y)^2 (D_x - 2D_y)} \sqrt{y + x}$$

$$= \frac{1}{(D_x - D_y)^2} \int [c - 2x + x]^{\frac{1}{2}} dx \text{ as corresponding to}$$

$$(D_x - 2D_y)z = 0,\ y + 2x = c$$

$$= \frac{1}{(D_x - D_y)^2} \sqrt{c - x}\ dx$$

$$= -\frac{2}{3} \frac{1}{(D_x - D_y)^2} (y + x)^{\frac{3}{2}} \text{ as } c = y + 2x$$

$$= -\frac{2}{3} \cdot \frac{1}{D_x - D_y} \int (c_1 - x + x)^{\frac{3}{2}} dx \text{ as corresponding to}$$

$$(D_x - D_y) = 0,\ y + x = c_1$$

$$= -\frac{2}{3} \cdot \frac{1}{D_x - D_y} c_1^{\frac{3}{2}} x$$

$$= -\frac{2}{3} \cdot \frac{1}{D_x - D_y} (y + x)^{\frac{3}{2}} x$$

$$= -\frac{2}{3} \int (c_2 - x + x)^{\frac{3}{2}} x\ dx$$

$$= -\frac{2}{3} \int c_2^{\frac{3}{2}} x\ dx$$

$$= -\frac{2}{3}c_2^{\frac{3}{2}}\frac{x^2}{2} = -\frac{1}{3}(x+y)^{\frac{3}{2}}x^2$$

$$\therefore z = \phi_1(y+2x) + \phi_2(y+x) + x\phi_3(y+x) - \frac{x^2}{3}(y+x)^{\frac{3}{2}}.$$

Example-11.12 Solve $(D_x^2 - 4D_y^2)z = \frac{4x}{y^2}$

Solution The auxiliary equation is $m^2 - 4 = 0$ giving $m = 2, -2$

$\therefore$ C. F $= \phi_1(y+2x) + \phi_2(y-3x)$

$$\text{P. I.} = \frac{1}{(D_x^2 - 4D_y^2)} \cdot \frac{4x}{y^2}$$

$$= \frac{1}{(D_x + 2D_y)(D_x - 2D_y)} \frac{x}{y^2}$$

$$= \frac{1}{(D_x + 2D_y)} \int \frac{4x}{(c-2x)^2} dx$$ as corresponding to

$$(D_x - 2D_y)z = 0,\ y + 2x = c$$

$$= \frac{1}{(D_x + 2D_y)} \int \left[-\frac{2}{(c-2x)} + \frac{2c}{(c-2x)^2} \right] dx$$

$$= \frac{1}{(D_x + 2D_y)} \left[\log(c-2x) + \frac{c}{c-2x} \right]$$

$$= \frac{1}{(D_x + 2D_y)} \left[\log y + \frac{y+2x}{y} \right]$$

$$= \int \left[\log(c'+2x) + \frac{c'+2x}{c'+2x} \right] dx$$ as corresponding to

$$(D_x + 2D_y)z = 0,\ y - 2x = c'$$

$$= \int \left[\log(c'+2x) + 2 - \frac{c'}{c'+2x} \right] dx$$

$$= x\log(c'+2x) - \int \frac{2x}{c'+2x} dx + 2x - \frac{1}{2}\log(c'+2x)$$

$$= x\log(c'+2x) - \int \frac{c'+2x-c'}{c'+2x} dx + 2x - \frac{1}{2}\log(c'+2x)$$

$$= x\log(c'+2x) - x + \frac{c'}{2}\log(c'+2x) + 2x - \frac{1}{2}\log(c'+2x)$$

$$= x\log y - x + \frac{y-2x}{2}\log y + 2x - \frac{1}{2}\log y$$

$$= \left(x + \frac{y-2x}{2} - \frac{1}{2} \right) \log y + x$$

General solution is

$$z = \phi_1(y+2x) + \phi_2(y-2x) + \left(x + \frac{y-2x}{2} - \frac{1}{2} \right) \log y + x$$

Non-homogeneous linear partial differential equation of higher order

An equation of the form $f(D_x, D_y)z = f(x, y)$

where $F(D_x, D_y)z = \left[A_0 D_x^n + A_1 D_x^{n-1} D_y + A_n D_x^n + B_0 D_y^{n-1} \right.$

$$\left. + B_1 D_x^{n-2} D_y + \cdots\cdots + B_{n-1} D_y^{n-1} + L_0 D_x + L_1 D_y + M \right] z$$

is called a non-homogeneous linear differential equation of order n and its complete solution consists of a complementary function and a particular integral. The problems are classified into two types.

(i) If $f(D_x, D_y)z$

can be expressed as a product of the linear factors of the form

$$(a_1 D_x + b_1 D_y + c_1)(a_2 D_x + b_2 D_y + c_2) \cdots\cdots (a_n D_x + b_n D_y + c_n)z$$

then the equation is said to be reducible.

(ii) If $f(D_x, D_y)z$

can not be factorised in linear factors of D_x and D_y then the equation is called irreducible equation. Linear factor such as $(D_x^2 + D_y)z = e^{x+y}$ is an example of irreducibl equation.

$$= e^{2x+y}\frac{1}{(D_x-2D_y+1)(D_x+D_y+8)}(x^2-y)$$

$$= \frac{1}{8}e^{2x+y}\frac{1}{(D_x-2D_y+1)}\left[1+\frac{D_x+D_y}{8}\right]^{-1}(x^2-y)$$

$$= \frac{1}{8}e^{2x+y}\frac{1}{(D_x-2D_y+1)}\left[1-\frac{D_x}{8}-\frac{D_y}{8}+\frac{D_x^2}{64}+\ldots\right](x^2-y)$$

$$= \frac{1}{8}e^{2x+y}\frac{1}{(D_x-2D_y+1)}\left[x^2-y-\frac{x}{4}+\frac{1}{8}+\frac{1}{32}\right]$$

$$= \frac{1}{8}e^{2x+y}(1+2D_y-D_x+D_x^2 \cdots\cdots)\left(x^2-\frac{x}{4}-y+\frac{5}{32}\right)$$

$$= \frac{1}{8}e^{2x+y}\left[x^2-\frac{x}{4}-y+\frac{5}{32}-2-2x+\frac{1}{4}+2\right]$$

$$= \frac{1}{8}e^{2x+y}\left(x^2-\frac{9x}{4}-y+\frac{5}{32}\right)$$

$$= \frac{1}{256}\left(32x^2-72x-32y+5\right)e^{2x+y}$$

General solution is

$$z = e^{-5x}\phi_1(y-x)+e^{-x}+\frac{1}{256}(32x^2-72x-32y+5)e^{2x+y}.$$

11.6 Homogeneous linear equation with variable coefficients

A partial differential equation of the form $F(xD_x, yD_y) = f(x, y)$ having variable coefficients can sometimes be reduced to equations with constant coefficients by suitable transformations. Let us change the independent variables x, y to u and v by the substitutions

$$x = e^u,\ y = e^v$$

or

$$u = \log x,\ v = \log y$$

$$D_x z = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} = \frac{1}{x}\frac{\partial z}{\partial y}$$

or

$$x\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \text{ or } xD_x z = D_u z$$

Again

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{1}{x}\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial u}\left(\frac{1}{x}\frac{\partial z}{\partial u}\right)\frac{\partial u}{dx}$$

$$= \frac{\partial}{\partial u}\left(\frac{1}{x}\frac{\partial z}{\partial u}\right)\frac{1}{x}$$

$$= \frac{1}{x}\left[\frac{1}{x}\frac{\partial^2 z}{\partial u^2} - \frac{\partial z}{\partial u}.\frac{1}{x^2}\frac{\partial x}{\partial u}\right]\frac{1}{x} = \frac{1}{x^2}\left(\frac{\partial^2 z}{\partial x^2} - \frac{\partial z}{\partial u}\right)$$

$$x^2\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial u^2} - \frac{\partial z}{\partial u} = D_u(D_u-1)z$$

$$\therefore \qquad x^2\frac{\partial^2}{\partial x^2} = D_u(D_u-1).$$

Similarly

$$x^3\frac{\partial^3}{\partial x^3} = D_u(D_u-1)(D_u-2)$$

Proceeding in a similar way we obtain

$$y\frac{\partial}{\partial y} = D_v$$

$$y^2\frac{\partial^2}{\partial y^2} = D_v(D_v-1)$$

$$y^3\frac{\partial^3}{\partial y^3} = D_v(D_v-1)(D_v-2) \text{ etc.}$$

Example 11.20 Solve $x^2\dfrac{\partial^2 z}{\partial x^2} - y^2\dfrac{\partial^2 z}{\partial y^2} = x^2 y.$

Solution The substitution $x = e^u$, $y = e^v$ transforms the given equation into

$$[D_u(D_u-1) - D_v(D_v-1)]z = e^{2u+v}$$

so that for any $a = a_i,\ b_i = \dfrac{3a_i^2 + a_i - 1}{2}$.

Thus $$C.F = \sum_{i=1}^{n} C_i e^{aix+biy}$$

where $$3a_r^2 - 2b_r^2 + a_r - 1 = 0$$

$$\text{P. I} = \frac{1}{3D_x^2 - 2D_y^2 + D_x - 1} 3e^{x+y}\sin(x+y)$$

$$= 3e^{x+y}\frac{1}{3(D_x+1)^2 - 2(D_y+1)^2 + (D_x+1) - 1}\sin(x+y)$$

$$= 3e^{x+y}\frac{1}{3D_x^2 - 2D_y^2 + 7D_x - 4D_y + 1}\sin(x+y)$$

$$= 3e^{x+y}\frac{1}{-3+2+7D_x - 4D_y + 1}\sin(x+y)$$

$$= 3e^{x+y}\frac{7D_x + 4D_y}{49D_x^2 - 16D_y^2}\sin(x+y)$$

$$= 3e^{x+y}\frac{11\cos(x+y)}{-49+16}$$

$$= 3e^{x+y}\frac{\cos(x+y)}{-3}$$

$$= -e^{x+y}\cos(x+y)$$

$\therefore$ The general solution

$$z = \sum_{r=1}^{\infty} c_r e^{a_r x + b_r y} - e^{x+y}\cos(x+y)$$

where $$3a_r^2 - 2b_r^2 + a_r - 1 = 0.$$

Example 11.18 Solve $\left(3D_x^2 - 2D_y^2 + D_x + 4\right)z = 7\tan(x+2y)$

Solution the equation is irreducible

Here $$f(a, b) = 3a^2 - 2b^2 + a + 4 = 0$$

sot that for any $a = ai, bi = \dfrac{3a_1^2 + a_1 + 4}{2}$

thus $$P.I = \frac{1}{3D_x^2 - 2D_y^2 + D_x + 4}.7\tan(x+2y)$$

Where $D_x = 1$, and $D_y = 2$, $3D_x^2 - 2D_y^2 + D_x + 4 = 0$.

It is a case of failure.

Thus $$P.I = 7x.\frac{1}{6D_x + 1}\tan(x+2y)$$

[multiplying numerator by x and differentiating the denominator w.r.t D_x]

$\therefore$ $$\text{P. I.} = \frac{7x}{7}\tan(x+2y)$$

$\therefore$ Complete solution is $z = \sum_{i=1}^{n} c_i e^{a_i x + b_i y} + x\tan(x+2y)$

where $$3a_r^2 - 2a_r^2 + a_r + 4 = 0$$

Example 11.19 Solve

$$(D_x^2 - D_x D_y - 2D_y^2 + 6D_x - 9D_y + 5)z = (x^2 - y)e^{2x+y}.$$

Solution For C. F. the equation may be written as

$$(D_x^2 - D_x D_y - 2D_y^2 + 6D_x - 9D_y + 5)z = 0$$

or $$(D_x - 2D_y + 1)(D_x + D_y + 5)z = 0$$

$$\therefore\ z_c = e^{-x}\phi_2(y+2x) + e^{-5x}\phi_1(y-x)$$

$$\text{P. I.} = \frac{1}{(D_x - 2D_y + 1)(D_x + D_y + 5)}.(x^2 - y)e^{2x+y}$$

$$=$$

$$e^{2x+y}\frac{1}{\{(D_x+2) - 2(D_y+1) + 1\}\{D_x + 2 + D_y + 1 + 5\}}(x^2 - y)$$

$$=\frac{1}{2}\frac{1}{(D_x+D_y)}\left[1+D_y-\frac{1}{2}D_x+D_y^2-D_xD_y+\frac{1}{4}D_x^2+\cdots\cdots\right]xy$$

$$=\frac{1}{2}\frac{1}{(D_x+D_y)}\left[xy+x-\frac{1}{2}y-1\right]$$

$$=\frac{1}{2}\frac{1}{D_x}\left(1+\frac{D_y}{D_x}\right)^{-1}\left(xy+x-\frac{1}{2}y-1\right)$$

$$=\frac{1}{2}\frac{1}{D_x}\left(1-\frac{D_y}{D_x}\right)\left(xy+x-\frac{1}{2}y-1\right)$$

$$=\frac{1}{2}\frac{1}{D_x}\left(xy+x-\frac{1}{2}y-1-\frac{x^2}{2}+\frac{x}{2}\right)$$

$$=\frac{1}{2}\frac{1}{D_x}\left(xy-\frac{1}{2}y-\frac{x^2}{2}+\frac{3x}{2}-1\right)$$

$$=\frac{1}{2}\left(\frac{x^2y}{2}-\frac{xy}{2}-\frac{x^3}{6}+\frac{3x^2}{4}-x\right)$$

$$=\frac{x}{24}\left(6xy-6y-2x^2+9x-12\right)$$

Hence the general solution is

$$z=\phi_1(y)+e^{-2x}\phi_2(2x+y)-\frac{1}{21}e^{3x+4y}+\frac{x}{24}(6xy-6y-2x^2+9x-12)$$

Example 11.16 Solve $(D_x^2+D_xD_y-6D_y^2)z=x^2\cos(x+y)$.

Solution The equation is $(D_x-2D_y)(D_x+3D_y)z=x^2\cos(x+y)$

$$\text{C. F.}=\phi_1(y+2x)+\phi_2(y-3x)$$

To evaluate the P. I. we take $\cos(x+y)$ as real part of $e^{i(x+y)}$

$$\text{P. I.}=\text{Real part of }\frac{1}{D_x^2+D_xD_y-6D_y^2}x^2e^{i(x+y)}$$

$$=\text{Real part of }e^{i(x+y)}\frac{1}{(D_x+i)^2+i(D_x+i)-6i^2}x^2$$

putting D_x+i for D_x and i for D_y.

$$=\text{Real part of }e^{i(x+y)}\frac{1}{D_x^2+3iD_x+4}x^2$$

-1

$$=\text{Real part of }\frac{e^{i(x+y)}}{4}\left[1+\frac{D_x^2+3iD_x}{4}\right]x^2$$

$$=\text{Real part of }\frac{e^{i(x+y)}}{4}\left[1-\frac{3iD_x}{4}-\frac{D_x^2}{4}-\frac{9D_x^2}{16}\right]x^2$$

$$=\text{Real part of }\frac{e^{i(x+y)}}{4}\left(x^2-\frac{3ix}{2}-\frac{13}{8}\right)$$

$$=\text{Real part of }\frac{\cos(x+y)+i\sin(x+y)}{4}\left(x^2-\frac{3ix}{2}-\frac{13}{8}\right)$$

$$=\frac{1}{4}\left\{\left(x^2-\frac{13}{8}\right)\cos(x+y)+\frac{3x}{2}\sin(x+y)\right\}$$

the general solution is

$$=z=\phi_1(y+2x)+\phi_2(y-3x)$$

$$+\frac{1}{4}\left\{\left(x^2-\frac{13}{8}\right)\cos(x+y)+\frac{3x}{2}\sin(x+y)\right\}$$

Remark In fact D_y should be replaced by D_y+i. But in the numerator D_y will be treated as dummy derivative since the differentiation will be carried out on x variable. So D_y is replaced by i alone.

Example 11.17 Solve $(3D_x^2-2D_y^2+D_x-1)z=3e^{x+y}\sin(x+y)$.

Solution The equation is irreducible.

Here, $f(a,b)=3a^2-2b^2+a-1=0$

11.4 Reducible equations

Complementary function. Let the equation be

$$F(D_x, D_y)z = (a_1D_x + b_1D_y + c_1)(a_2D_x + b_2D_y + c_2)$$

or $$(a_rD_x + b_rD_y + c_r) = f(x, y)$$

where $a_r, b_r, c_n (n = 1, n)$ are constants.

Let us consider the equation

$$(a_1D_x + b_1D_y + c_1)z = 0$$

or $$a_1p + b_1q + c_1z = 0$$

The auxiliary equations are $\frac{dx}{a_1} = \frac{dy}{b_1} = \frac{dz}{-c_1z}$

From the first two relations we get, $a_1y - b_1x = A$

Case-1: If $a_1 \neq 0$ then from $\frac{dx}{a_1} = \frac{dz}{-c_1z}$

we get $$\frac{dz}{z} = -\frac{c_1}{a}dx$$

or $$\log z = -\frac{c_1x}{a} + \log B$$

or $$z = Be^{-\frac{c_1x}{a_1}}$$

or $$ze^{\frac{c_1x}{a_1}} = B$$

$\therefore$ The general solution is

$$z = e^{-\frac{c_1x}{a_1}}\phi(a_1y - b_1x) \tag{11.14}$$

Case-2: If $b_1 \neq 0$ then from $\frac{dy}{b_1} = \frac{dz}{-c_1z}$

we get $$ze^{\frac{c_1y}{b_1}} = c$$

$\therefore$ The general solution may be expressed as

$$z = e^{-\frac{c_1y}{b_1}}\phi(a_1y - b_1x) \tag{11.15}$$

Thus if no two factors are linearly dependent (i.e. no factor is a multiple of another) the general solution of

$$f(D_x, D_y)z = 0$$

is the sum of n arbitrary functions of the types mentioned above.

In the case of repeated factors, for example

$$(D_x - mD_y - a)^3 z = 0$$

we have $z = e^{ax}\phi_1(y + mx) + xe^{ax}\phi_2(y + mx) + x^2e^{ax}\phi_3(y + mx)$

Case-3: If $c_1 = 0$, the equation becomes homogeneous linear equation and its solution is obtained if we put $c_1 = 0$ in equation (11.14) or (11.15) which reduces to $z = \phi(a_1y - b_1x)$.

Expt: $\frac{\partial^4 z}{\partial x^4} + \frac{\partial^4 z}{\partial y^4} = 2\frac{\partial^4 z}{\partial x^2 \partial y^2}$

11.5 Irreducible equation

If complementary function.

When $F(D_x, D_y)z$ can not be factorized into linear factors we apply for complementary function the trial method given below:

Let us consider the linear equation with constant coefficients

$$F(D_x, D_y)z = 0 \tag{11.16}$$

Since $$D_x^r D_y^s (Ce^{ax+by}) = ca^r b^s e^{ax+by} \tag{11.17}$$

were a, b, c are constants. The trial substitution

$$z = ce^{ax+by}$$

in (11.16) gives $$cf(a, b)e^{ax+by}$$

is a solution of (11.16) provided

$$f(a, b) = 0 \tag{11.18}$$

Exp: Find a P.I. of the eq^n:

$$(D_x^2 - D_y)z = 2y - x^2$$

Sol^n:

$$z_{P.I.} = \frac{1}{D_x^2 - D_y}\, 2y - x^2$$

$$= -\frac{1}{D_y}\left(1 - \frac{D_x^2}{D_y}\right)^{-1}(2y - x^2)$$

$$= -\frac{1}{D_y}\left(1 + \frac{D_x^2}{D_y} + \frac{D_x^4}{D_y^2} + \dots\right)(2y - x^2)$$

$$= \left(-\frac{1}{D_y} - \frac{D_x^2}{D_y^2} - \frac{D_x^4}{D_y^3} - \dots\right)(2y - x^2)$$

$$= \underline{-y^2 + x^2 y} - \frac{1}{D_y^2}(-2)$$

$$= -y^2 + x^2 y + y^2$$

$$\therefore z_{P.I.} = x^2 y$$

$$z = z_{cf} + z_{P.I}.$$

#

When $f(x,y)$ is of the form e^{ax+by} then a P.I. is made up of terms of the form $\frac{1}{F(a,b)} e^{ax+by}$ if $F(a,b) \neq 0$.

Exp: Find a P.I. of $(D_x^2 - D_y)z = e^{2x+y}$

$$F(D_x, D_y) = D_x^2 - D_y$$

$$F(a,b) = F(2,1) = 2^2 - 1 = 3$$

$$\therefore \text{P.I. of } z = \frac{1}{3} e^{2x+y}$$

#

When $F(a,b) = 0 \Rightarrow$

Exp: Find a P.I. of $(D_x^2 - D_y)z = e^{x+y}$.

Sol^n: $F(a,b) = 1^2 - 1 = 0 \Rightarrow F(D_x, D_y) \sim F(D_x + a, D_y + b)$

Consider, $z = w e^{x+y}$

We know, $F(D_x, D_y)z = ce^{ax+by}$

Here,

$$F(D_x+1, D_y+1)\, w e^{x+y} = e^{x+y}$$

$$\cancel{\Rightarrow (D_x^2 - D_y)w = 1}$$

$$[(D_x+1)^2 - (D_y+1)]\, w = 1$$

$$(D_x^2 + 2D_x - D_y)\, w = 1$$

$$w = \frac{1}{D_x^2 + 2D_x - D_y}$$

$$= \frac{1}{2D_x\left(1 + \frac{D_x^2 - D_y}{2D_x}\right)}$$

$$= \frac{1}{2D_x} \cdot \left(1 + \frac{D_x^2 - D_y}{2D_x}\right)^{-1}$$

$$= \frac{1}{2D_x} \cdot 1$$

$$= \frac{1}{2}x$$

Again,

$$w = \frac{1}{-D_y\left(1 - \frac{D_x^2 + 2D_x}{D_y}\right)}$$

$$= -\frac{1}{D_y} \cdot 1 = -y$$

$$\boxed{\text{Since } z = w e^{x+y} = \tfrac{1}{2} x e^{x+y} \text{ and } y e^{x+y} \text{ are P.I.s.}}$$

A. E. is $(D_u^2 - D_v^2 - D_u + D_v)z = 0$

or $(D_u - D_v)(D_u - D_v - 1)z = 0$

$$C.F = \phi_1(v+u) + e^u\phi_2(v-u)$$
$$= \phi_1(\log y + \log x) + x\phi_2(\log y - \log x)$$
$$= f_1(xy) + xf_2\left(\frac{y}{x}\right)$$

$$\text{P. I.} = \frac{1}{(D_u - D_v)(D_u + D_v - 1)}.e^{2u+v}$$
$$= \frac{1}{2}e^{2u+v} = \frac{1}{2}x^2y$$

General solution is $z = C.F + P.F. = f_1(xy) + xf_2\left(\frac{y}{x}\right) + \frac{1}{2}x^2y.$

Example 11.21 Solve

$$\left(x^2\frac{\partial^2 z}{\partial x^2} - 4y^2\frac{\partial^2 z}{\partial y^2} - 4y\frac{\partial z}{\partial y} - 1\right)z = x^2y^3\log y.$$

Solution Substituting $x = e^u$, $y = e^v$ and denoting

$$\frac{\partial}{\partial u} \text{ and } \frac{\partial}{\partial v}$$

by D_u and D_v respectively the given equation reduces to

$$[D_u(D_u - 1) - 4D_v(D_v - 1) - 4D_v - 1]z = ve^{2u+3v}$$

C. F. is $z = \sum_{i=1}^{\infty} c_i e^{a_i u + b_i v}$

$$\text{P. I.} = \frac{1}{D_u^2 - 4D_v^2 - D_u - 1}ve^{2u+3v}$$
$$= e^{2u+3v}\frac{1}{(D_u+2)^2 - 4(D_v+3)^2 - (D_u+2) - 1}v$$
$$= e^{2u+3v}\frac{1}{D_u^2 - 4D_v^2 + 3D_u - 24D_v - 35}v$$

$$= e^{2u+3v}\left[-\frac{v}{35} + \frac{24}{35^2}\right] = -\frac{1}{1225}e^{2u+3v}(35v - 24)$$

$$\therefore \quad z = \sum_{i=1}^{\infty} c_i e^{a_i u + b_i v} - \frac{1}{1225}x^3y^3(35\log y - 24)$$

where $a_i^2 - 4b_i^2 - a_1' - 1 = 0$.

Example 11.22 Solve $yt - q = xy$.

Solution The given equation can be written as

$$y\frac{\partial^2 z}{\partial y^2} - \frac{\partial z}{\partial y} = xy$$

or
$$y^2\frac{\partial^2 z}{\partial y^2} - y\frac{\partial z}{\partial y} = xy^2$$

Substituting $x = e^u,\ y = e^v$

we get $[D_v(D_v - 1) - D_v]z = e^{u+2v}$

or $D_v(D_v - 2)z = e^{u+2v}$

$$\text{C.F.} = f_1(\log x) + y^2 f_2(\log x)$$
$$= \phi_1(x) + y^2\phi_2(x)$$

$$\text{P. I.} = \frac{1}{D_v^2 - 2D_v}e^{u+2v}$$
$$= \frac{v}{2D_v - 2}e^{u+2v}$$
$$= \frac{v}{2}e^{u+2v} = \frac{1}{2}xy^2\cos y.$$

General solution is $z = \phi_1(x) + y^2\phi_2(x) + \frac{1}{2}xy^2\log y.$

Example 11.23 Solve

$$\left(x^2\frac{\partial^2 z}{\partial x^2} - 2xy\frac{\partial^2 y}{\partial x\partial y} - 3y^2\frac{\partial^2 z}{\partial y^2} + \frac{\partial y}{\partial x} - 3\frac{\partial z}{\partial y}\right) = x^2y\sin(\log x^2)$$

$$= \frac{1}{n(n-1)}\left[e^{nu} + \frac{n}{2}e^{(n-2)u+2v} + \frac{\frac{n}{2}\left(\frac{n}{2}-1\right)}{1.2}e^{(n-4)u+4v}\right]$$

$$= \frac{1}{n(n-1)}e^{nu}\left[1+e^{2(v-u)}\right]^{\frac{n}{2}}$$

$$= \frac{\left(e^{2u}+e^{2v}\right)^{\frac{n}{2}}}{n(n-1)} = \frac{\left(x^2+y^2\right)^{\frac{n}{2}}}{n(n-1)}$$

G. S. is $z = \phi\left(\frac{y}{x}\right) + x\phi_2\left(\frac{y}{x}\right) + \frac{\left(x^2+y^2\right)^{\frac{n}{2}}}{n(n-1)}$

Example 11.25 Solve

$$x^2\left(\frac{\partial^2 z}{\partial x^2} - xy\frac{\partial^2 z}{\partial x\partial y} - 2y\frac{\partial^2 z}{\partial y^2} + x\frac{\partial z}{\partial x} - 2y\frac{\partial x}{\partial y}\right)z = \log\left(\frac{y}{x}\right) - \frac{1}{2}.$$

Solution $x = e^u$ and $y = e^v$ reduces the given equation to

$$\left[D_u(D_u-1) - D_uD_v - 2D_v(D_v-1) + D_u - 2D_v\right]z = v - u - \frac{1}{2}$$

$$\left(D_u^2 - D_uD_v - 2D_v^2\right)z = v - u - \frac{1}{2}$$

$$(D_u - 2D_v)(D_u + D_v)z = v - u - \frac{1}{2}$$

$$z_c = f_1(v+2u) + f_2(v-u)$$

$$= \phi_1(x^2y) + \phi_2\left(\frac{y}{x}\right)$$

$$\text{P. I} = \frac{1}{(D_u^2 - D_uD_v - 2D_v^2)}\left(v - u - \frac{1}{2}\right)$$

$$= \frac{1}{D_u^2 - D_uD_v - 2D_v^2}(v-u) - \frac{1}{2}\frac{1}{(D_u^2 - D_uD_v - 2D_v^2)}$$

$$= \frac{1}{D_u^2}\left[1 - \left(\frac{D_v}{D_u} + \frac{2D_v^2}{D_u^2}\right)\right]^{-1}(v-u) - \frac{1}{2}\cdot\frac{u^2}{2}$$

$$= \frac{1}{D_u^2}\left[1 + \frac{D_v}{D_u} + \frac{2D_v^2}{D_u^2}\right](v-u) - \frac{u^2}{4}$$

$$= \frac{1}{D_u^2}[v - u + u] - \frac{u^2}{4}$$

$$= \frac{vu^2}{2} - \frac{u^2}{4}$$

$$= \frac{1}{2}(\log x)^2\log y - \frac{1}{4}(\log x)^2$$

Therefore the complete solution is

$$z = \phi_1(x^2y) + \phi_2\left(\frac{y}{x}\right) + \frac{1}{2}(\log x)^2\log y - \frac{1}{4}(\log x)^2.$$

Example 11.26 Solve $\frac{1}{x^2}\frac{\partial^2 z}{\partial x^2} - \frac{1}{x^3}\frac{\partial z}{\partial x} = \frac{1}{y^2}\frac{\partial^2 z}{\partial y^2} - \frac{1}{y^3}\frac{\partial z}{\partial y}$

Solution Put $\frac{1}{2}x^2 = u$ and $\frac{1}{2}y^2 = v$

$$\therefore\ x\partial x = \partial u \quad ; \quad y\partial y = \partial v$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} = \frac{1}{x}\frac{\partial z}{\partial x}$$

$$\frac{\partial^2 z}{\partial u^2} = \frac{\partial}{\partial u}\left(\frac{\partial z}{\partial u}\right) = \frac{\partial}{\partial x}\left(\frac{1}{x}\frac{\partial z}{\partial x}\right)\frac{\partial x}{\partial u}$$

$$= \left[-\frac{1}{x^2}\frac{\partial z}{\partial x} + \frac{1}{x}\frac{\partial^2 z}{\partial x^2}\right]\frac{1}{x}$$

$$= -\frac{1}{x^3}\frac{\partial z}{\partial x} + \frac{1}{x^2}\frac{\partial^2 z}{\partial x^2}$$

Similarly $$\frac{1}{y^2}\frac{\partial^2 z}{\partial y^2}-\frac{1}{y^3}\frac{\partial z}{\partial y}=\frac{\partial^2 z}{\partial y^2}$$

or $$\frac{\partial^2 z}{\partial u^2}-\frac{\partial^2 z}{\partial v^2}=0$$

or $$(D_u-D_v)(D_u+D_v)z=0$$

Therefore the solution is $z=f_1(v+u)+f_2(v-u)$

$$=f_1\left(\frac{y^2+x^2}{2}\right)+f_2\left(\frac{y^2-x^2}{2}\right)$$

$$=\phi_1(y^2+x^2)+\phi_2(y^2-x^2)$$

Example 11.27 Find a surface satisfying

$$2x^2r-5xys+2y^2t+2(px+qy)=0$$

and touching the hyperbolic paraboloid $z=x^2-y^2$ along its section by the plane $y=1$.

Solution The given equation can be written as

$$2x^2\frac{\partial^2 z}{\partial x^2}-5xy\frac{\partial^2 z}{\partial x\partial y}+2y^2\frac{\partial^2 z}{\partial y^2}+2\left(x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}\right)=0$$

$$x=e^u \text{ and } y=e^v$$

reduces the equation to

$$[2D_u(D_u-1)-5D_uD_v+2D_v(D_v-1)+2(D_u+D_v)z]=0$$

or $$(2D_u-D_v)(D_u-2D_v)z=0$$

Hence the solution is

$$z=f_1(2v+u)+f_2(v+2u)$$

$$=f_1(2\log y+\log x)+f_2(\log y+2\log x)$$

$$=f_1(\log xy^2)+f_2(\log x^2y)$$

$$=\phi_1(xy^2)+\phi_2(x^2y)$$

The other surface is $z=x^2-y^2$

Since the two surfaces touch each other along the section by $y=1$, the values of p and q for two surfaces must be equal at $y=1$.

Now, $$p=\frac{\partial z}{\partial x}=y^2\phi_1'(xy^2)+2xy\phi_2'(x^2y)=2x \quad (11.19)$$

and $$q=\frac{\partial z}{\partial y}=2xy\phi_1'(xy^2)+x^2\phi_2'(x^2y)=2y \quad (11.20)$$

For $y=1$, equation (11.19) and equation (11.20) reduces to

$$\phi_1'(x)+2x\phi_2'(x^2)=2x \quad (11.21)$$

and $$2x\phi_1'(x)+x^2\phi_2'(x^2)=-2 \quad (11.22)$$

Solving (11.21) and (11.22) we get

$$\phi_1'(x)=-\frac{2x}{3}-\frac{4}{3x} \quad (11.23)$$

and $$\phi_2'(x^2)=\frac{2}{3x^2}+\frac{4}{3} \quad (11.24)$$

Integrating (11.23) we get $\phi_1(x)=-\frac{x^2}{3}-\frac{4}{3}\log x$ (11.25)

Integrating (11.24) w.r.t. x^2 we obtain

$$\phi_2(x^2)=\frac{2}{3}\log x^2+\frac{4}{3}x^2$$

$$=\frac{4}{3}\log x+\frac{4}{3}x^2 \quad (11.26)$$

Replacing x by xy^2 in (11.25) we get

$$\phi_1(xy^2)=-\frac{1}{3}x^2y^4-\frac{4}{3}\log xy^2$$

$$=-\frac{1}{3}x^2y^4-\frac{8}{3}\log y-\frac{4}{3}\log x$$

Replacing x^2 by x^2y, in (11.26) we get

$$\phi_2(x^2y)=\frac{4}{3}\log(x\sqrt{y})+\frac{4}{3}x^2y$$

$$=\frac{4}{3}\log x+\frac{2}{3}\log y+\frac{4}{3}x^2y \quad (11.27)$$

$$X = A_1 \cos\lambda x + B_1 \sin\lambda x \text{ and } T = C_1 e^{-\frac{\lambda^2}{4}t} \text{ respectively.}$$

Thus a solution of the partial differential equation is,

$$u(x, t) = XT = (A_1\cos\lambda x + B_1\sin\lambda x)C_1 e^{-\frac{\lambda^2 t}{4}}$$

$$= e^{-\frac{\lambda^2}{4}t}(A\cos\lambda x + B\sin\lambda x) \qquad (11.32)$$

where $A_1C_1 = A$ and $B_1C_1 = B$.

Since $u(0, t) = 0$ $\quad \therefore e^{-\frac{\lambda^2}{4}t}(A+0) = 0$

or, $Ae^{-\frac{\lambda^2}{4}t} = 0$ $\quad \therefore A = 0$, since $e^{-\frac{\lambda^2}{4}t} \neq 0$

Thus from (11.32), we have $u(x, t) = Be^{-\frac{\lambda^2}{4}t}\sin\lambda x \qquad (11.33)$

since $u(2, t) = 0$, $\quad \therefore Be^{-\frac{\lambda^2}{4}t}\sin 2\lambda = 0$

So we must have $\sin 2\lambda = 0 \quad (B \neq 0)$

Or, $2\lambda = n\pi$ where $n = 0, \pm 1, \pm 2, \ldots\ldots \quad \therefore \lambda = \frac{n\pi}{2}$.

Thus a solution is $u(x, t) = Be^{-\frac{n^2\pi^2 t}{16}}\sin\frac{n\pi x}{2} \qquad (11.34)$

Now by the principle of superposition, we have,

$$u(x, t) = B_1 e^{-\frac{n_1^2\pi^2 t}{16}}\sin\frac{n_1\pi x}{2} + B_2 e^{-\frac{n_2^2\pi^2 t}{16}}\sin\frac{n_2\pi x}{2}$$

$$+ B_3 e^{-\frac{n_3^2\pi^2 t}{16}}\sin\frac{n_3\pi x}{2} \qquad (11.35)$$

Using the last boundary condition, we have

$$u(x, 0) = B_1\sin\frac{n_1\pi x}{2} + B_2\sin\frac{n_2\pi x}{2} + B_3\sin\frac{n_3\pi x}{2}$$

$$= 2\sin\frac{\pi x}{2} - \sin\pi x + 4\sin 2\pi x$$

which is possible if and only if $B_1 = 2,\ n_1 = 1;$

$$B_2 = 1,\ n_2 = 2;\quad B_3 = 4,\ n_3 = 4.$$

Putting these values is (11.35), we get

$$u(x, t) = 2e^{-\frac{\pi^2 t}{16}}\sin\frac{\pi x}{2} - e^{-\frac{\pi^2 t}{4}}\sin\pi x + 4e^{-\pi^2 t}\sin 2\pi x$$

which is the required solution of the given heat conduction problem.

Test your knowledge

Solve

1. $(D_x^2 + 6D_xD_y + 9D_y^2)z = 0.$
2. $(D_x^3 - 2D_x^2D_y + 9D_y^2)z = 0.$
3. $(D_x^3 - 2D_x^2D_y)z = 3x^2y.$
4. $(D_x^2 + 2D_xD_y + D_y^2)z = x^2 + xy + y^2.$
5. $(D_x^2 + 5D_xD_y + 6D_y^2)z = e^{x-y}.$
6. $(2D_x^2 - D_xD_y - 3D_y^2)z = 5e^{x-y}.$
7. $(D_x - 2D_y)^2(D_x + 3D_y)z = e^{2x+y}.$
8. $(D_x^3 - 7D_xD_y^2 - 6D_y^3)z = \sin(x+2y).$
9. $(D_x^2 - 3D_xD_y + 2D_y^2)z = \cos(x+2y).$
10. $(D_x^3 - 4D_x^2D_y + 4D_xD_y^2)z = \sin(2x+y).$
11. $(D_x^3 + D_x^2D_y - D_xD_y^2) = e^x\cos 2y.$
12. $(D_x^2 - 5D_xD_y + 6D_y^2)z = e^{x-y}xy.$
13. $(4D_x^2 - 4D_xD_y + D_y^2)z = \log(x+2y).$